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ML: Assignment - 8

Q.1.

→ ① Clustering results change because different distance metrics define similarity differently.

ⓐ Euclidean distance measures absolute difference in feature values. Hence, features with larger scales dominate the clustering, making clusters CGPA-oriented.

ⓑ Cosine distance measure the angle between ~~co~~ vectors ~~...~~. Thus, it groups students based on similar skill profiles.

② Cosine distance is better when the relative pattern of features is more important than their absolute values, or when data is not normalized and features have different scales.

Q.2.

→ ① MST clustering identifies non-convex shapes because it relies on local connectivity rather than global centroids. By connecting points via their shortest neighbours, the algorithm forms a tree based on proximity. A cluster is defined as a connected component. Thus, it can follow any arbitrary, winding "chain" as long as internal points remain close.

② The primary weakness is sensitivity to noise, known as the "Chaining Effect". A single outlier positioned between two distinct, dense clusters can act as a "bridge". If the edges connecting this outlier are shorter than the threshold for removal, the MST will fail to separate the groups, merging them into a single, poorly defined cluster.

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Q. 3.

→ ① DBSCAN forms clusters based on density using two parameters:

ⓐ ϵ : minimum radius of neighbourhood.

ⓑ MinPts : minimum no. of points in an ϵ neighbourhood of that point.

These parameters control the definition of a cluster as:

① Increasing ϵ increases neighbourhood size, resulting in more core points ~~and fewer clusters~~ as more points satisfy the MinPts requirement. ~~Decreasing~~ This leads to merging of smaller clusters into larger ones, thus decreasing the total cluster count.

② A single ϵ value may not work very well for datasets with varying density. As dense groups of high-performing students and sparse groups of average students may not be correctly identified, leading to either merged clusters or excessive noise.

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